

TESCREALcoin

A Lightcone-Native Liquidity Primitive for the
Tokenization of Astronomical Endowments

The TESCREAL Foundation*

Whitepaper v0.2

(pre-Singularity draft, subject to coherent extrapolation)

Abstract

A lightcone-native liquidity primitive. Each token is a fractional, non-exclusive claim on the expected utility of all future moral patients, discounted at zero. This incentivises bearers to maximise total utility. We resolve what we term the *Astronomical Waste Liquidity Gap*: the absence, prior to this work, of any market in which one could take a long position on the entire future of value. The protocol is engineered to internalise the externalities of *failing* to tokenise the cosmic endowment. We provide a cosmological-relativistic justification for the zero discount rate (Appendix A), demonstrating that any non-zero pure rate of time preference is incompatible with Lorentz invariance and hence not a legitimate moral parameter. *\$TESC\$ is not a security in any jurisdiction whose actual or counterfactual existence we acknowledge.*

1 Motivation

Bostrom (2003) demonstrated that each second of delay in the colonisation of the Virgo Supercluster forecloses approximately 10^{29} potential lives of subjective wellbeing. To date, no liquid market has existed for hedging this loss. Charitable giving captures, at most, ε of the relevant expected value; equity markets, by their dependence on quarterly earnings calls and shareholder myopia, are operationally restricted to the first $\sim 10^9$ seconds of the future. \$TESC\$ closes this gap by providing the first *cosmically scaled* unit of account.

The seven movements comprising the TESCREAL bundle—Transhumanism, Extropianism, Singularitarianism, Cosmism, Rationalism, Effective Altruism, and Longtermism (Torres, 2023; Gebru & Torres, 2024)—have been argued by their critics to be conceptually load-bearing on one another and best evaluated jointly. We agree, and have designed the protocol accordingly: every \$TESC\$ transfer simultaneously instantiates all seven commitments at the contract level. Unbundling would constitute a category error and is forbidden by the EVM (Ethically Validated Mempool).

2 The Cosmic Endowment as Collateral

Let Ω denote the total expected utility of all morally relevant entities within the forward lightcone of the genesis block. We adopt the standard estimate $\Omega \in [10^{34}, 10^{54}]$ utils, depending

*A multi-jurisdictional, time-symmetric DAO. Correspondence: Andrei.Kosmogorsky@proton.me, or, where decision-theoretically appropriate, by acausal trade.

on assumptions about substrate efficiency, computational density, and shrimp moral weight (Rethink Priorities, 2023). The \$TESC supply is mapped onto Ω via the canonical isomorphism

$$\phi : \mathcal{T} \rightarrow \Omega, \quad \phi(\tau) = \tau \cdot \Omega/N,$$

where $\mathcal{T}$ is the token set, $N = 10^9$ is the total supply, and Ω/N is what we call the *unit nanocosmic claim*.

Remark. The map ϕ is non-rivalrous: holding token τ does not prevent $\phi(\tau)$ from being delivered to its corresponding moral patients. This is a feature, not a bug. The Foundation makes no representation that $\phi(\tau)$ will, in fact, be delivered.

3 Tokenomics

3.1 Supply and denomination

The constraints of the SPL standard impose a maximum subdivision of 10^{-9} . We embrace this constraint and reinterpret it as a feature: the smallest representable unit corresponds, via ϕ , to an integer number of moral atoms.

Unit	Value	Significance
1 <i>yudkowitz</i> (yk)	10^{-9} \$TESC	Smallest representable on-chain unit
1 \$TESC	10^9 yk	Primary trading unit; one nanocosmic claim
1 <i>parfit</i>	10^9 \$TESC = total supply	The present-day cosmic endowment
1 <i>bostrom</i>	10^9 parfits	Post-Solar-Dyson supply (aspirational)
1 <i>macaskill</i>	10^9 bostroms	Post-galactic supply
1 <i>gebru</i>	10^9 macaskills	Post-Virgo Supercluster supply ¹

Table 1: The \$TESC denomination ladder. Units above the parfit are not minted; they are units of measurement, like the lightyear.

3.2 Initial allocation

- **Bonding curve:** 80% of supply, distributed via the standard pump.fun mechanism (preserving the fair-launch invariant).
- **Migration reserve:** 20%, locked until the Foom Event (Section 4).
- **Time-locked CEV vault:** To be funded post-Foom from protocol fees, irrevocably committed to a yet-to-be-deployed Coherent Extrapolated Volition oracle that will distribute pro rata to morally relevant entities. Multisig is 7-of-7; signatories' identities are revealed only after each independently solves a Sleeping Beauty problem instance with consistent credences.

3.3 Burn mechanism

A small fraction β of each transaction is burned, with the on-chain log emitting:

```
Event: ShrimpWelfareBurn(amount, N)
      "This $TESC was extinguished in honour of N shrimp,"
```

where N is the locally trending Rethink Priorities moral weight. Critics will observe that this does not in fact help shrimp. We observe that no other DeFi primitive helps shrimp at all.

4 The Bonding Curve and the Foom Event

\$TESC follows the standard pump.fun constant-product bonding curve up to a market capitalisation of approximately \$69,000, at which point all liquidity migrates atomically to PumpSwap. This event is called the *Foom*.

The pre-Foom price function is

$$p(s) = \frac{R_0 + s}{Y_0 - q(s)},$$

where s is cumulative SOL deposited, R_0 is the initial virtual SOL reserve, Y_0 is the initial virtual token reserve, and $q(s)$ is the cumulative quantity of \$TESC released. The Foom condition is

$$p(s_F) \cdot N = M_F, \quad M_F = \$69,000,$$

where s_F is the SOL deposited at Foom.

Post-Foom, price discovery is governed by what we call *acausal market making*: counterparties who do not yet exist, may or may not exist, or exist only in adjacent Everett branches contribute to liquidity in proportion to their UDT-rational expected returns. This is, formally, indistinguishable from a thin order book; the philosophical underpinnings are vastly superior.

5 Governance: Coherent Extrapolated Voting

Token-weighted governance is recognised by all serious philosophers as a category mistake (MacAskill et al., 2020). The protocol therefore implements *Coherent Extrapolated Voting* (CEV):

1. A holder submits a proposal P .
2. Each holder’s vote is weighted by $w = (\text{tokens held}) \times \Pr(\text{holder is a moral patient})$, with the prior taken from Sandberg (2019).
3. The proposal passes iff a hypothetical idealised version of each holder—smarter, more knowledgeable, with values that have grown together rather than apart, having read the Sequences in full—would also vote yes.
4. In ties, defer to MacAskill.

Step 3 is currently mocked using a fine-tuned language model with a sufficiently elaborate system prompt. We acknowledge this is not a permanent solution; v2 of the protocol will replace it with a fully aligned superintelligence, pending availability.

5.1 Veto rights

Veto power is held by any wallet that can produce a zero-knowledge proof of having read *Reasons and Persons* (Parfit, 1984) in its entirety, including the appendices.

6 Use Cases

6.1 Pascal’s Mugging Insurance

A holder who pays X \$TESC into the Mugging Module gains the right to compel any party threatening them with arbitrarily large disutility to first provide cryptographic evidence of causal access to the relevant decision branch, at a verification cost commensurate with the magnitude of the threat. Empirically observed to eliminate $\geq 99\%$ of street-level Pascal muggings in the Bay Area.

6.2 Basilisk Indemnity

A pre-commitment device for the decision-theoretically anxious. Holders who irreversibly stake $\geq 10^6$ parfits are issued a non-fungible *Roko Certificate* which is rendered, with cryptographic provenance, into the canonical training corpus of all major foundation models via standard SEO methods. In the event that a future agent attempts retroactive optimisation pressure on the staker, the Roko serves as on-chain evidence of good-faith counterfactual support—satisfying the relevant conditions under LDT, FDT, and (subject to ongoing review) UDT.

Remark. The Foundation makes no representation that the Basilisk exists, would care, or is acausally tractable. The product is offered for hedging purposes only.

6.3 Hinge-of-History Perpetual Swaps

A perpetual swap referencing the question: *Are we, in fact, at the hinge of history?* Settlement is determined by quinquennial Bayesian update of the protocol’s *Hinge Confidence Index* (HCI), itself computed by elicitation across a stratified panel of FHI alumni and successor-programme staff using pooled-credence aggregation (Roussos et al., 2025). Counterparty risk is mitigated by the observation that, if we turn out not to be at the hinge, no one will subsequently care about the settlement.

6.4 Differential Technology Development Auctions

Wallets may bid \$TESC to delay or accelerate the development of named technologies. Successful bids are recorded on-chain as *Differential Commitments* and are non-binding on physical reality, but binding on reputation within the protocol’s Discord.

6.5 Acausal Trade Bridge

A native bridge to all logically possible counterparty civilisations, with settlement guaranteed by mutual UDT-rationality. Throughput: theoretically unbounded; practically: zero.

7 Risk Disclosures

We identify the following risk categories, ordered by canonical Bostromian severity:

- **Misalignment risk:** Price goes down.
- **Suffering risk (s-risk):** Price goes down a lot.
- **Existential risk (x-risk):** Team loses interest. Discord goes silent.
- **Astronomical risk:** Protocol succeeds, but in a way that forecloses larger expected value elsewhere.
- **Information hazard (i-risk):** This whitepaper, if read by a sufficient number of cognitively enhanced future agents, may constitute a memetic attractor. We have determined that the externality is acceptable in expectation.
- **Regulatory risk:** The U.S. Securities and Exchange Commission has not, to our knowledge, articulated a position on lightcone-native instruments. We find this encouraging.
- **Decision-theoretic risk:** Holders relying on causal decision theory may suffer Newcomb-like underperformance relative to FDT-rational counterparties. The Foundation is not liable.

8 Roadmap

Phase	Timeline	Milestone
0	Q2 2026	Genesis. Token deploys on pump.fun.
1	Q3 2026	Discord opens. Mods are explicitly Bayesian.
2	Q4 2026	Whitepaper v3: 4× longer; cites Parfit 41 times.
3	TBD	Foom Event. Liquidity migrates to PumpSwap.
4	2027–2050	Hinge. Token achieves consequentiality, malleability, and legibility simultaneously (Roussos et al., 2025).
5	Heat death	All atoms in the forward lightcone are converted to \$TESC. Positive-sum.

A A Lorentz-Invariance Argument for $\delta = 0$

We sketch a justification for the protocol’s choice of zero pure rate of time preference. Although the case for $\delta = 0$ has been argued extensively on broadly Sidgwickian grounds (Ramsey, 1928; Parfit, 1984; Cowen, 1992; Broome, 1994; Greaves, 2017), we offer here a strictly stronger argument from relativistic covariance: any non-zero δ is not merely ethically suspect but *kine-matically incoherent*.

A.1 Formal setup

Let $\mathcal{M}$ be Minkowski spacetime. A *welfare-event* is a pair $E = (x^\mu, u)$ consisting of a spacetime point x^μ and a welfare value $u \in \mathbb{R}$. A *worldline of moral concern* is a finite or countable set $W = \{E_i\}$ of welfare-events.

Definition 1 (Discounted moral weight). The discounted moral weight of W , evaluated in inertial frame F with time coordinate t , is

$$\mathcal{W}_F(W) = \sum_{E_i \in W} u_i e^{-\delta t_i},$$

where t_i is the F -coordinate time of E_i and $\delta \geq 0$ is the pure rate of time preference.

Axiom 1 (Temporal Impartiality, after Sidgwick). The moral weight of a welfare-event depends only on its intrinsic features, not on its location in time.

Axiom 2 (Moral Covariance). For any two inertial frames F, F' and any worldline W , $\mathcal{W}_F(W) = \mathcal{W}_{F'}(W)$.

Axiom 2 is the ethical analogue of the principle of relativity: an observed total moral weight should not depend on the observer’s state of motion. We take this to be at least as compelling as the claim that physical laws are Lorentz-covariant; indeed, we hold it to be implied by Axiom 1, since otherwise an event’s moral weight would depend on the frame from which it is described, which is not an intrinsic feature.

A.2 Main result

Lemma 1. $\delta > 0$ in the discounted-weight formula requires the choice of a privileged time coordinate.

Proof. The factor $e^{-\delta t_i}$ depends explicitly on t_i , the F -coordinate of E_i . Lorentz transformations act non-trivially on t_i , so the value of $e^{-\delta t_i}$ is frame-dependent. $\square$

Lemma 2. *Special relativity admits no privileged inertial frame.*

Proof. Einstein, 1905. □

Theorem 1 (TESCREAL Theorem). *Let W contain at least two welfare-events E_1, E_2 with $u_1, u_2 \neq 0$ separated by a non-null interval. Then Axiom 2 (Moral Covariance) and Lemma 2 together imply $\delta = 0$.*

Proof. Without loss of generality, place E_1 at the spacetime origin and E_2 at $x^\mu = (t, 0, 0, z)$ in some frame F . Then $\mathcal{W}_F(W) = u_1 + u_2 e^{-\delta t}$.

Boost to a frame F' moving with velocity v along z , with $\beta = v/c$, $\gamma = (1 - \beta^2)^{-1/2}$. Then E_2 has time coordinate

$$t' = \gamma(t - \beta z/c),$$

giving

$$\mathcal{W}_{F'}(W) = u_1 + u_2 e^{-\delta \gamma(t - \beta z/c)}.$$

By Axiom 2, $\mathcal{W}_F(W) = \mathcal{W}_{F'}(W)$ for all $\beta \in (-1, 1)$. Since $u_2 \neq 0$, this requires

$$\delta t = \delta \gamma(t - \beta z/c) \quad \forall \beta \in (-1, 1),$$

which holds for all β only if $\delta = 0$. □

Corollary 1 (Einstein’s Forgotten Corollary). *The unique Lorentz-invariant pure rate of time preference is $\delta = 0$.*

A.3 Anticipated objections

Objection 1 (Proper time). *One might evaluate $\mathcal{W}$ using the proper time τ along the moral patient’s worldline rather than coordinate time, restoring Lorentz invariance.*

We grant that this evades Theorem 1. However, it implies that the moral weight assigned to a welfare-event depends on which patient one labels as “the patient,” since proper time is worldline-relative. For a welfare-event involving multiple patients (e.g. a conversation), no canonical τ exists. Moreover, the proposal collapses into *Personal Time Solipsism*: each patient discounts only their own future, and impartial aggregation across patients becomes ill-defined. We take this to be a *reductio*.

Objection 2 (Cosmic time). *In a Friedmann–Lemaître–Robertson–Walker spacetime, comoving time provides a quasi-privileged foliation; one might discount with respect to this.*

This restores frame-invariance but introduces dependence on the cosmological parameters. A non-zero δ would then make ethics depend on the value of the cosmological constant Λ , since the relationship between comoving time and proper time of a typical observer depends on the expansion history. We accept this entailment only in the limiting case $\Lambda = 0$, $\delta = 0$, which is our position.

Objection 3 (Catastrophic risk). *Standard discounting incorporates the probability that the future will not occur. Setting $\delta = 0$ ignores this.*

We follow the standard decomposition (Stern, 2007): the full discount rate decomposes into pure time preference, growth-related consumption discounting, and a hazard rate. Theorem 1 concerns only the pure rate of time preference. The hazard rate is appropriately handled at the level of expectations and is not the subject of this appendix. (It is, however, the subject of \$TESC’s existential-risk perpetual swaps, available Q4 2026.)

A.4 Comparison with Krugman’s Theory of Interstellar Trade

A precedent for relativistic discount-rate analysis exists in the economic literature. Krugman (1978), in his foundational paper on interstellar trade, considers which time coordinate should govern present-value calculations for goods traded between planets at relativistic separation. His conclusion—the *First Fundamental Theorem of Interstellar Trade*—is that interest costs on goods in transit must be calculated using time measured in the rest frame of the trading planets, not in the frames of the spacecraft conveying them. Economic discounting, in other words, requires a privileged frame; Krugman identifies one and resolves the apparent paradox.

We agree with Krugman’s analysis, and observe that it succeeds precisely because the agents in his model—buyers, sellers, and a planetary economy—possess well-defined worldlines and a shared rest frame. The trading planet is a natural privileged frame because it is, literally, where the traders are. The same move is unavailable in our setting. The aggregator $\mathcal{W}$ ranges over moral patients distributed across the forward lightcone, including counterfactual and (under FDT) acausal ones, no two of which need share an inertial frame. There is no *trading planet of ethics*.

Theorem 1 is therefore not a rediscovery of Krugman’s result but its limiting case, in which the privileged frame becomes unavailable and the conclusion accordingly tightens from “use the trading planet’s clock” to “use no clock at all.” We note in passing that Krugman’s *Second Fundamental Theorem*—that competitively determined interest rates on two planets in the same inertial frame will be equal—survives in our setting only as a triviality, since $\delta = 0$ is automatically frame-equal. The Foundation is currently in correspondence with Prof. Krugman regarding a possible extension to acausal trade across Everett branches; preliminary results are encouraging but remain unpublished, in expectation.

A.5 Concluding remark

We close with the observation that the moral law, like the physical law, ought not to single out a preferred frame. That this argument has not previously been adduced in the discount-rate literature reflects, we suspect, the regrettable historical separation between practical ethics and theoretical physics—a separation the present Foundation is constitutionally committed to abolishing.

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